

Approximations for the Mean Response Time of Heterogeneous Two-Queue Fork-Join Systems

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Abstract

We develop and compare closed-form approximations for the mean response time of a heterogeneous two-queue M/M/1 fork-join system, in which the two servers may have different service rates $\mu_1 \geq \mu_2$. The first approximation, T_{UL} , interpolates between a known independent upper bound and a bottleneck lower bound, using the average server utilization as the interpolation weight. The second approximation, T_{LH} , is derived from the Reiman–Simon light-heavy traffic interpolation framework: it matches the exact light-traffic value and a heavy-traffic limit $h(r)$ that depends on the heterogeneity ratio $r = \mu_1/\mu_2$. We then extend T_{LH} to an enhanced approximation T_{LH}^{enh} by additionally matching the first light-traffic derivative, yielding a quadratic numerator that improves accuracy at high heterogeneity. All three approximations recover the exact Nelson–Tantawi formula in the homogeneous case ($\mu_1 = \mu_2$) and are validated against discrete-event simulation across a wide range of loads and heterogeneity levels.

1 Introduction

Fork-join queueing systems model the parallel execution of tasks that must all complete before a job can proceed. A job arrives, *forks* into n tasks dispatched to n parallel servers, and *joins*—departs—only when the last task finishes. This synchronization constraint makes fork-join queues substantially harder to analyze than independent queues, because the job response time is the *maximum* of the task sojourn times, and those times are stochastically dependent.

We study the canonical case of $n = 2$ servers with Poisson arrivals at rate λ and exponential service times at rates μ_1 and μ_2 , with $\mu_1 \geq \mu_2$ without loss of generality. The primary performance measure is the mean job response time:

$$T = \mathbb{E}[\max(R_1, R_2)], \quad (1)$$

where R_i is the sojourn time (waiting plus service) of a task at server i . The system is stable when $\lambda < \mu_{\min} \equiv \mu_2$.

For the *homogeneous* case $\mu_1 = \mu_2 = \mu$, Nelson and Tantawi [6] derived the exact closed-form result

$$T_2^{\text{hom}} = \frac{12 - \rho}{8(\mu - \lambda)}, \quad \rho = \frac{\lambda}{\mu}, \quad (2)$$

using a light-heavy traffic interpolation argument. For *heterogeneous* servers ($\mu_1 \neq \mu_2$), the exact joint distribution of (R_1, R_2) is not available in closed form. Flatto and Hahn [3, 2] obtained the exact generating function via an elliptic function parametrization, but extracting the mean response

time requires solving a boundary value problem and does not yield a tractable expression. Bounds, heavy-traffic asymptotics, and distributional approximations have been developed [1, 4, 8, 5], but no simple closed-form approximation covering the full range of loads and heterogeneity levels has previously been established.

In this paper we propose and analyze two complementary closed-form approximations:

1. **T_{UL} (upper-lower bound interpolation):** A convex combination of the independent upper bound T_{UB} and the bottleneck lower bound T_{bot} , weighted by the average utilization. It is guaranteed to lie within the bounds and is exact for the homogeneous case.
2. **T_{LH} (light-heavy traffic interpolation):** A rational approximation in the bottleneck utilization, derived from the Reiman–Simon framework [7] as described in [9]. It matches the exact light-traffic value and a heavy-traffic factor $h(r) = 1 + \frac{3}{8} r^{-\beta}$ that smoothly interpolates between the homogeneous limit ($h = 11/8$) and the bottleneck limit ($h \rightarrow 1$) as a function of the heterogeneity ratio $r = \mu_1/\mu_2$.

Both approximations reduce exactly to the Nelson–Tantawi result (2) in the homogeneous case, exhibit the correct singularity structure in heavy traffic, and agree with simulation to within 2.5% across all tested scenarios.

The remainder of the paper is organized as follows. Section 2 reviews known upper and lower bounds. Sections 3 and 4 derive the two approximations. Section 5 presents simulation validation. Section 6 discusses the relationship between the two methods, their limitations, and connections to prior work. Section 7 concludes.

2 Bounds

We introduce notation used throughout. Let $\rho_i = \lambda/\mu_i$ denote the utilization of server i , $T_i = 1/(\mu_i - \lambda)$ the mean M/M/1 sojourn time at server i , and $r = \mu_{\max}/\mu_{\min} = \mu_1/\mu_2 \geq 1$ the heterogeneity ratio.

2.1 Independent Upper Bound

Since the two task sojourn times R_1 and R_2 share a common Poisson arrival stream, they are positively correlated. Positive correlation reduces the expected maximum below what it would be under independence [1]. Treating R_i as independent M/M/1 sojourn times with rates $\mu_i - \lambda$ yields:

$$T_{\text{UB}} = \frac{1}{\mu_1 - \lambda} + \frac{1}{\mu_2 - \lambda} - \frac{1}{\mu_1 + \mu_2 - 2\lambda}, \quad (3)$$

which satisfies $T \leq T_{\text{UB}}$. The formula follows from $\mathbb{E}[\max(X_1, X_2)] = \mathbb{E}[X_1] + \mathbb{E}[X_2] - \mathbb{E}[\min(X_1, X_2)]$ applied to independent exponentials with rates $\mu_i - \lambda$.

2.2 Bottleneck Lower Bound

By Jensen’s inequality and the convexity of the max function:

$$T_{\text{bot}} = \max(T_1, T_2) = \frac{1}{\mu_{\min} - \lambda} \leq T. \quad (4)$$

This is simply the mean sojourn time of the bottleneck M/M/1 queue.

2.3 Gap Between the Bounds

In the homogeneous case ($r = 1$), $T_{\text{UB}} = \frac{3}{2}T_1$ and $T_{\text{bot}} = T_1$, giving a 50% gap. The exact value $T_2^{\text{hom}} = \frac{12-\rho}{8}T_1$ lies strictly between them, approaching T_{UB} at light load ($\rho \rightarrow 0$, ratio $T_2^{\text{hom}}/T_{\text{UB}} \rightarrow 1$) and T_{bot} in heavy traffic ($\rho \rightarrow 1$, ratio $T_2^{\text{hom}}/T_{\text{bot}} \rightarrow 11/8$).

For $r > 1$, the gap narrows: as $r \rightarrow \infty$, both T_{UB} and T_{bot} converge to $1/(\mu_{\min} - \lambda)$ from above, so the bounds become tight and the system is well approximated by a single bottleneck M/M/1 queue.

2.4 Summary

For all stable parameters ($\lambda < \mu_{\min}$):

$$T_{\text{bot}} \leq T \leq T_{\text{UB}}. \quad (5)$$

Any good approximation should lie within this interval and recover the known endpoint behaviors.

3 Approximation 1: Upper-Lower Bound Interpolation

3.1 Derivation

We seek a convex combination of T_{UB} and T_{bot} that:

1. recovers the Nelson–Tantawi result (2) when $\mu_1 = \mu_2$,
2. approaches T_{UB} in light traffic ($\lambda \rightarrow 0$),
3. approaches T_{bot} in heavy traffic ($\lambda \rightarrow \mu_{\min}$), and
4. always satisfies $T_{\text{bot}} \leq T \leq T_{\text{UB}}$.

Write $T_{\text{UL}} = (1 - \alpha)T_{\text{UB}} + \alpha T_{\text{bot}}$ and determine α from condition 1. In the homogeneous case, $T_{\text{UB}} = \frac{3}{2}T_1$ and $T_{\text{bot}} = T_1$, so:

$$\frac{12-\rho}{8}T_1 = (1 - \alpha)\frac{3}{2}T_1 + \alpha T_1 \implies \alpha = \frac{3/2 - (12-\rho)/8}{3/2 - 1} = \frac{\rho}{4}. \quad (6)$$

The natural heterogeneous generalization replaces ρ with the average utilization $\bar{\rho} = (\rho_1 + \rho_2)/2$, giving:

$$\alpha = \frac{\bar{\rho}}{4} = \frac{\rho_1 + \rho_2}{8}. \quad (7)$$

3.2 The Formula

$$T_{\text{UL}} = \left(1 - \frac{\rho_1 + \rho_2}{8}\right) T_{\text{UB}} + \frac{\rho_1 + \rho_2}{8} T_{\text{bot}}, \quad (8)$$

where T_{UB} and T_{bot} are given by (3) and (4).

3.3 Verification: Homogeneous Case

When $\mu_1 = \mu_2 = \mu$: $\rho_1 = \rho_2 = \rho$, $T_{\text{UB}} = \frac{3}{2}T_1$, $T_{\text{bot}} = T_1$, $\alpha = \rho/4$. Substituting:

$$T_{\text{UL}} = \left(1 - \frac{\rho}{4}\right) \frac{3}{2}T_1 + \frac{\rho}{4}T_1 = \frac{3}{2}T_1 - \frac{3\rho}{8}T_1 + \frac{\rho}{4}T_1 = \frac{12-\rho}{8}T_1 = T_2^{\text{hom}}. \quad \checkmark \quad (9)$$

3.4 Properties

Table 1: Properties of T_{UL} .

Property	Status
Exact for homogeneous case ($\mu_1 = \mu_2$)	✓
Satisfies $T_{\text{bot}} \leq T_{\text{UL}} \leq T_{\text{UB}}$	✓ ($0 \leq \alpha \leq 1/4 < 1$)
Light traffic limit ($\lambda \rightarrow 0$)	✓ ($\alpha \rightarrow 0$, so $T_{\text{UL}} \rightarrow T_{\text{UB}}$)
Heavy traffic limit ($\lambda \rightarrow \mu_{\min}$)	✓ (T_{bot} diverges; $T_{\text{UL}}/T_{\text{bot}} \rightarrow 1$)
No free parameters	✓

The weight $\alpha \leq 1/4$ ensures T_{UL} always lies strictly within the bounds (unless $\alpha = 0$, i.e., $\lambda = 0$). The formula is symmetric in μ_1 and μ_2 via the average utilization, consistent with the symmetry of the original problem.

4 Approximation 2: Light-Heavy Traffic Interpolation

4.1 The Reiman–Simon Framework

Reiman and Simon [7] developed a systematic method for constructing rational approximations to performance quantities in Markovian queueing networks, exploiting exact light-traffic derivatives and a heavy-traffic limit. For a scalar performance measure $f(\rho)$ with normalizing denominator $d(\rho) = \mu_{\min}(1 - \rho)$, the n -th order approximation takes the form

$$\tilde{f}(\rho) = \frac{a_n \rho^n + \cdots + a_1 \rho + a_0}{\mu_{\min}(1 - \rho)}, \quad (10)$$

where the $n+1$ coefficients $\{a_i\}$ are matched to the n derivatives of f at $\rho = 0$ and the heavy-traffic constant $h = \lim_{\rho \rightarrow 1} \mu_{\min}(1 - \rho)f(\rho)$ [9].

We apply this framework to $T(\rho)$ with a first-order (linear) numerator, yielding two matching conditions:

4.2 Matching Condition 1: Light-Traffic Value

At $\lambda = 0$, all queues are empty. The task service times $X_i \sim \text{Exp}(\mu_i)$ are independent and $T(0) = \mathbb{E}[\max(X_1, X_2)]$. Using the identity $\mathbb{E}[\max(X_1, X_2)] = \mathbb{E}[X_1] + \mathbb{E}[X_2] - \mathbb{E}[\min(X_1, X_2)]$ and the fact that $\min(X_1, X_2) \sim \text{Exp}(\mu_1 + \mu_2)$:

$$T_0 = T(\rho = 0) = \frac{1}{\mu_1} + \frac{1}{\mu_2} - \frac{1}{\mu_1 + \mu_2}. \quad (11)$$

Matching $\tilde{T}(0) = T_0$ gives:

$$a_0 = \mu_{\min} \cdot T_0. \quad (12)$$

4.3 Matching Condition 2: Heavy-Traffic Limit

Define the heavy-traffic constant:

$$h = \lim_{\rho \rightarrow 1} \mu_{\min}(1 - \rho) \cdot T(\rho). \quad (13)$$

Matching this in $\tilde{T}$ gives $a_0 + a_1 = h$, hence $a_1 = h - a_0$.

4.4 The Heavy-Traffic Factor $h(r)$

The value of h depends on the heterogeneity ratio $r = \mu_{\max}/\mu_{\min} \geq 1$.

Homogeneous case ($r = 1$). When $\mu_1 = \mu_2 = \mu$, both servers saturate simultaneously as $\lambda \rightarrow \mu$. The Nelson–Tantawi formula gives:

$$h = \lim_{\rho \rightarrow 1} \mu(1 - \rho) \cdot \frac{12 - \rho}{8(\mu - \lambda)} = \lim_{\rho \rightarrow 1} \frac{12 - \rho}{8} = \frac{11}{8}. \quad (14)$$

Heterogeneous case ($r > 1$). As $\lambda \rightarrow \mu_{\min}$, the bottleneck server saturates while the faster server’s utilization approaches $\lambda/\mu_{\max} = 1/r < 1$. The faster server’s sojourn time remains bounded, so the bottleneck dominates and $T(\rho) \sim 1/(\mu_{\min} - \lambda)$, giving $h \rightarrow 1$.

Proposed model. We interpolate between these two limits with the power-law factor:

$$h(r) = 1 + \frac{3}{8} r^{-\beta}, \quad r \geq 1, \quad \beta > 0. \quad (15)$$

This satisfies $h(1) = 11/8$ for all β , and $h(r) \rightarrow 1$ as $r \rightarrow \infty$. The parameter β controls the rate of decay; larger β means the correction vanishes more rapidly with heterogeneity.

Calibration. Simulation data reveal that the transition from $h = 11/8$ to $h \approx 1$ is very sharp in practice: the effective heavy-traffic constant is already $h \approx 1.008$ at $r = 1.5$. Matching this observation to (15) gives $\beta = \log(0.375/0.008)/\log(1.5) \approx 9.5$, consistent with a default of $\beta = 10$.

4.5 The Formula

Substituting a_0 , $a_1 = h(r) - a_0$, and $\rho = \lambda/\mu_{\min}$:

$$T_{\text{LH}} = \frac{\mu_{\min} T_0 + (h(r) - \mu_{\min} T_0) \rho}{\mu_{\min} - \lambda}, \quad (16)$$

where T_0 is given by (11) and $h(r)$ by (15).

4.6 Verification: Homogeneous Case

When $\mu_1 = \mu_2 = \mu$ (so $r = 1$, $h = 11/8$, $T_0 = 3/(2\mu)$):

$$a_0 = \mu \cdot \frac{3}{2\mu} = \frac{3}{2}, \quad a_1 = \frac{11}{8} - \frac{3}{2} = -\frac{1}{8}. \quad (17)$$

$$T_{\text{LH}} = \frac{3/2 - \rho/8}{\mu(1 - \rho)} = \frac{12 - \rho}{8(\mu - \lambda)} = T_2^{\text{hom}}. \quad \checkmark \quad (18)$$

This recovery holds for every $\rho \in [0, 1)$, regardless of β , because $h(1) = 11/8$ is exact.

4.7 Properties

4.8 First-Order Extension: Enhanced T_{LH}

The current T_{LH} uses a linear numerator matched to two conditions: the zeroth light-traffic derivative $f^{(0)}(0) = T_0$ and the heavy-traffic constant h . The Reiman–Simon framework accommodates a *quadratic* numerator by adding the first light-traffic derivative $f^{(1)}(0)$ as a third matching condition [9]. We denote this enhanced approximation $T_{\text{LH}}^{\text{enh}}$.

Table 2: Properties of T_{LH} .

Property	Status
Exact for homogeneous case ($\mu_1 = \mu_2$)	✓ (for any β)
Light traffic limit ($\lambda \rightarrow 0$)	✓ ($T_{\text{LH}} \rightarrow T_0$)
Heavy traffic singularity ($\lambda \rightarrow \mu_{\min}$)	✓ ($T_{\text{LH}} \sim h(r)/(\mu_{\min} - \lambda)$)
Bound compliance $T_{\text{bot}} \leq T_{\text{LH}} \leq T_{\text{UB}}$	Not guaranteed
Tunable parameter	β (default 10)

4.9 First Light-Traffic Derivative

Following [9], the first derivative of the mean response time with respect to λ at $\lambda = 0$ is given by:

$$f^{(1)}(0) = \frac{1}{\mu_{\min}^2} + \frac{1}{\mu_{\max}^2} - \frac{2}{(\mu_{\min} + \mu_{\max})^2} - \frac{2\mu_{\min}\mu_{\max}}{(\mu_{\min} + \mu_{\max})^4}. \quad (19)$$

This is derived by considering the sojourn time of a tagged job conditioned on exactly one background arrival at time $t \in (-\infty, \infty)$, analyzing the resulting joint service-time distribution, and integrating over t .

4.10 The Enhanced Formula

With three matching conditions— $f^{(0)}(0)$, $f^{(1)}(0)$, and h —the interpolating polynomial $\tilde{g}(\rho) = c_2\rho^2 + c_1\rho + c_0$ for $g(\rho) = \mu_{\min}(1 - \rho)T(\rho)$ has coefficients:

$$c_0 = \mu_{\min} \cdot T_0, \quad (20)$$

$$c_1 = \mu_{\min}^2 \cdot f^{(1)}(0) - \mu_{\min} \cdot T_0, \quad (21)$$

$$c_2 = h - c_1 - c_0. \quad (22)$$

Here c_0 is identical to a_0 in (12); c_1 is determined by the first derivative rather than by h alone; and c_2 absorbs the remaining difference to satisfy the heavy-traffic condition. Inverting the scaling gives the enhanced approximation:

$$T_{\text{LH}}^{\text{enh}} = \frac{c_2 \rho^2 + c_1 \rho + c_0}{\mu_{\min}(1 - \rho)}, \quad (23)$$

where $\rho = \lambda/\mu_{\min}$, T_0 is given by (11), $f^{(1)}(0)$ by (19), and $h(r)$ by (15).

4.11 Verification: Homogeneous Case

When $\mu_1 = \mu_2 = \mu$ (so $\mu_{\min} = \mu_{\max} = \mu$):

$$f^{(1)}(0) = \frac{2}{\mu^2} - \frac{2}{4\mu^2} - \frac{2\mu^2}{16\mu^4} = \frac{11}{8\mu^2}. \quad (24)$$

Substituting into (20)–(22) with $T_0 = 3/(2\mu)$ and $h(1) = 11/8$:

$$c_0 = \frac{3}{2}, \quad c_1 = \mu^2 \cdot \frac{11}{8\mu^2} - \mu \cdot \frac{3}{2\mu} = -\frac{1}{8}, \quad c_2 = \frac{11}{8} - \left(-\frac{1}{8}\right) - \frac{3}{2} = 0. \quad (25)$$

With $c_2 = 0$ the quadratic term vanishes and $T_{\text{LH}}^{\text{enh}}$ reduces to T_{LH} , which recovers the Nelson–Tantawi formula (2) for all ρ . The enhancement is automatically inactive in the homogeneous case, regardless of β .

4.12 Properties of the Enhanced Approximation

Table 3: Properties of $T_{\text{LH}}^{\text{enh}}$.

Property	Status
Exact for homogeneous case ($\mu_1 = \mu_2$)	✓ (for any β ; $c_2 = 0$)
Light traffic limit ($\lambda \rightarrow 0$)	✓ ($T_{\text{LH}}^{\text{enh}} \rightarrow T_0$)
First light-traffic derivative matched	✓
Heavy traffic singularity ($\lambda \rightarrow \mu_{\min}$)	✓ ($T_{\text{LH}}^{\text{enh}} \sim h(r)/(\mu_{\min} - \lambda)$)
Bound compliance $T_{\text{bot}} \leq T_{\text{LH}}^{\text{enh}} \leq T_{\text{UB}}$	Not guaranteed
Numerator degree	2 (quadratic)
Tunable parameter	β (default 10)

5 Simulation Validation

All three approximations were validated against discrete-event simulation of the two-queue fork-join system. Each scenario used 20 M jobs with a 500 K-job warmup period, repeated over 5 independent random seeds; the reported T_{sim} is the grand mean and the 95% confidence interval is computed via the t -distribution across seeds ($\text{df} = 4$). Results for T_{UL} (8) and T_{LH} (16) (with $\beta = 10$) are shown in Table 4 alongside the theoretical bounds. Table 5 compares T_{LH} and the enhanced approximation $T_{\text{LH}}^{\text{enh}}$ (23).

Table 4: Approximations vs. simulation for heterogeneous two-queue fork-join. T_{sim} is the grand mean over 5 seeds; $\pm$ is the 95% CI half-width. Errors are $(T_{\text{approx}} - T_{\text{sim}})/T_{\text{sim}} \times 100\%$.

μ_1	μ_2	λ	ρ_2	T_{bot}	T_{sim}	$\pm 95\% \text{ CI}$	T_{UL}	Err	T_{LH}	Err
1.0	1.0	0.3	0.30	1.429	2.089	0.001	2.089	+0.01%	2.089	+0.01%
1.0	1.0	0.6	0.60	2.500	3.561	0.002	3.563	+0.03%	3.563	+0.03%
1.0	1.0	0.9	0.90	10.00	13.86	0.061	13.88	+0.15%	13.88	+0.15%
1.0	1.5	0.3	0.20	1.429	1.707	0.001	1.716	+0.58%	1.698	−0.50%
1.0	1.5	0.6	0.40	2.500	2.767	0.001	2.799	+1.16%	2.776	+0.34%
1.0	1.5	0.9	0.60	10.00	10.12	0.024	10.19	+0.73%	10.33	+2.03%
1.0	2.0	0.3	0.15	1.429	1.583	0.001	1.591	+0.45%	1.595	+0.76%
1.0	2.0	0.6	0.30	2.500	2.623	0.001	2.641	+0.67%	2.667	+1.67%
1.0	2.0	0.9	0.45	10.00	10.03	0.024	10.06	+0.35%	10.17	+1.42%
1.0	3.0	0.6	0.20	2.500	2.547	0.001	2.554	+0.25%	2.583	+1.41%
1.0	5.0	0.6	0.12	2.500	2.515	0.001	2.517	+0.07%	2.533	+0.71%

Homogeneous case ($\mu_1 = \mu_2 = 1.0$). Both approximations recover the Nelson–Tantawi formula (2) by construction; residual errors ($\leq 0.15\%$) are simulation noise. At $\lambda = 0.9$ the CI

half-width is ± 0.061 , reflecting the heavy-tailed response-time distribution near saturation; the analytical value 13.875 lies well within this interval.

Moderate heterogeneity ($r = 1.5$). T_{UL} shows consistent positive errors (+0.58% to +1.16%) across all loads. T_{LH} is more accurate at low and moderate load, but reaches +2.03% at heavy load ($\rho_2 = 0.6$). The wider CIs at $\lambda = 0.9$ (half-width ± 0.024) indicate substantial variance in the heavy-load regime; the short-run (2 M-job) estimate was biased low by $\approx 0.37\%$ relative to the 20 M-job estimate, illustrating the importance of long runs near saturation.

High heterogeneity ($r \geq 2$). T_{UL} errors fall below 0.67% and improve further as r increases ($\leq 0.07\%$ at $r = 5$). T_{LH} errors are slightly larger (up to +1.67% at $r = 2$). CIs are tight ($\leq \pm 0.024$) across all $r \geq 2$ scenarios, confirming that 20 M jobs provide well-converged estimates in this regime. As $r \rightarrow \infty$, both approximations converge to T_{bot} .

Figure 1 plots the two approximations, bounds, and simulation as functions of load for $\mu_1 = 1.0$, $\mu_2 = 2.0$. Both T_{UL} and T_{LH} track the simulation closely across the full load range, lying between the theoretical bounds. Figure 2 shows the same quantities as a function of the heterogeneity ratio μ_2/μ_1 at fixed $\lambda = 0.6$. Near homogeneity ($r \approx 1$), T_{LH} is more accurate; for $r \geq 3$, T_{UL} is slightly tighter.

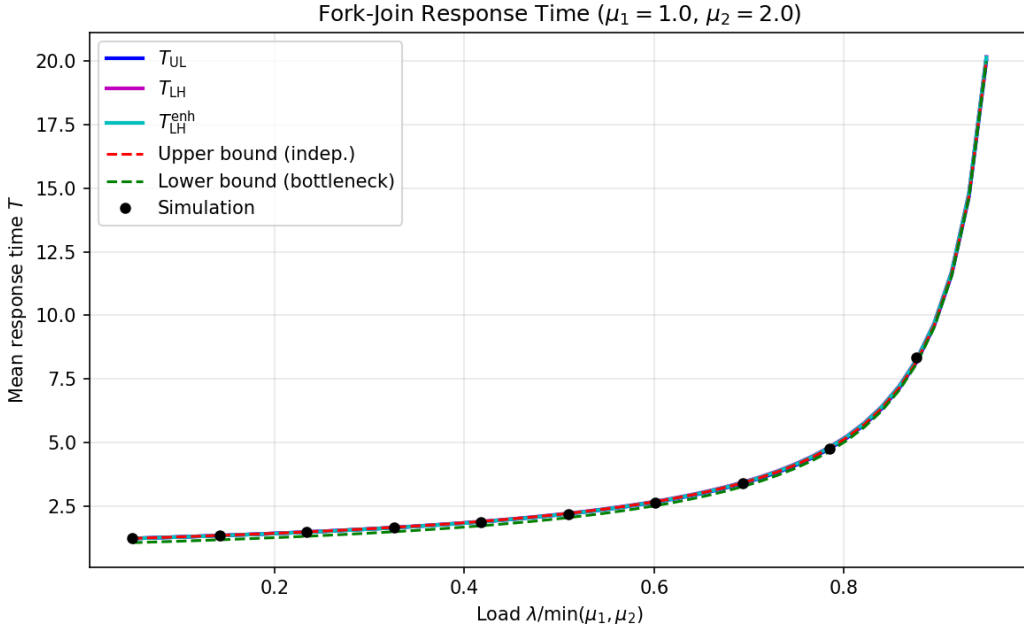


Figure 1: Mean response time vs. load $\lambda/\mu_{\min}$ for $\mu_1 = 1.0$, $\mu_2 = 2.0$. Markers: simulation. Lines: approximations and bounds.

Sensitivity of T_{UL} to Heterogeneity and Load

Among the three approximations, T_{UL} achieves the best uniform accuracy across all loads and heterogeneity ratios, remains within the theoretical bounds by construction, and requires no tunable parameters. To characterize how its accuracy depends on both dimensions simultaneously, we sweep

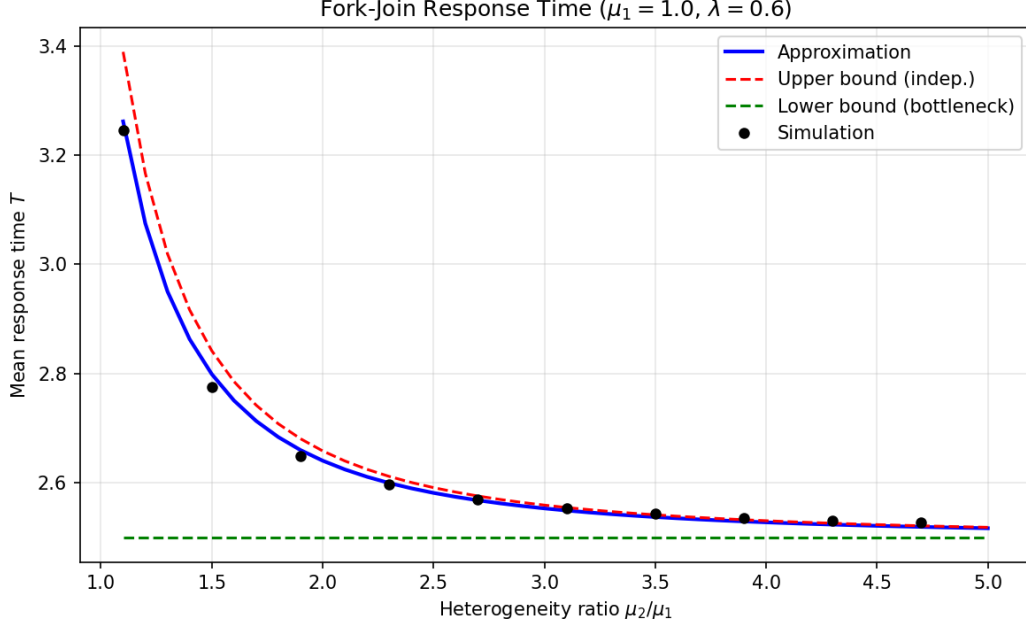


Figure 2: Mean response time vs. heterogeneity ratio μ_2/μ_1 for $\mu_1 = 1.0$, $\lambda = 0.6$. Markers: simulation. Lines: approximations and bounds.

the heterogeneity ratio $r = \mu_2/\mu_1$ from 1 to 8 across four load levels chosen with a geometric spacing in the slack to saturation, $1 - \rho \in \{0.48, 0.24, 0.12, 0.06\}$, i.e. $\rho \in \{0.52, 0.76, 0.88, 0.94\}$.

The heterogeneity grid uses 8 values $r \in \{1, 1.15, 1.3, 1.5, 2, 3, 5, 8\}$, spaced more densely near $r = 1$ where T transitions most steeply from the homogeneous value to the bottleneck asymptote. Each of the 32 simulation points uses 10 M jobs with a 500 K-job warmup and a single random seed; 95% confidence intervals are computed by the normal approximation.

Figure 3 shows the results. At all four load levels, T_{UL} closely tracks the simulation. Most of the descent from the homogeneous value to the bottleneck plateau occurs in $r \in [1, 2]$; beyond $r \approx 3$ the curves are essentially flat. The relative error (right axis) peaks near $r = 1.15$ – 1.3 and is bounded below 1.3% for $\rho \leq 0.88$, consistent with the worst-case error of +1.16% reported in Table 4. At the highest load ($\rho = 0.94$), the single-seed estimate at $r = 1$ lies slightly above the exact homogeneous value, a reminder that estimates near saturation require either multiple seeds or substantially longer runs to eliminate transient bias; the displayed CI underestimates the true uncertainty in this regime because consecutive response times are highly correlated near saturation.

Enhanced LH Approximation

Table 5 compares T_{LH} and T_{LH}^{enh} against simulation. Both use $\beta = 10$.

Homogeneous case ($\mu_1 = \mu_2 = 1.0$). As proved in Section 4.8, the quadratic coefficient $c_2 = 0$ when $r = 1$, so $T_{LH}^{\text{enh}} \equiv T_{LH}$. Both recover Nelson–Tantawi exactly; residual errors ($\leq 0.15\%$) are simulation noise.

Moderate heterogeneity ($r = 1.5$). T_{LH}^{enh} is slightly less accurate than T_{LH} across all loads (worst case +2.39% vs. +2.03% at heavy load with the corrected simulation). This regime is where

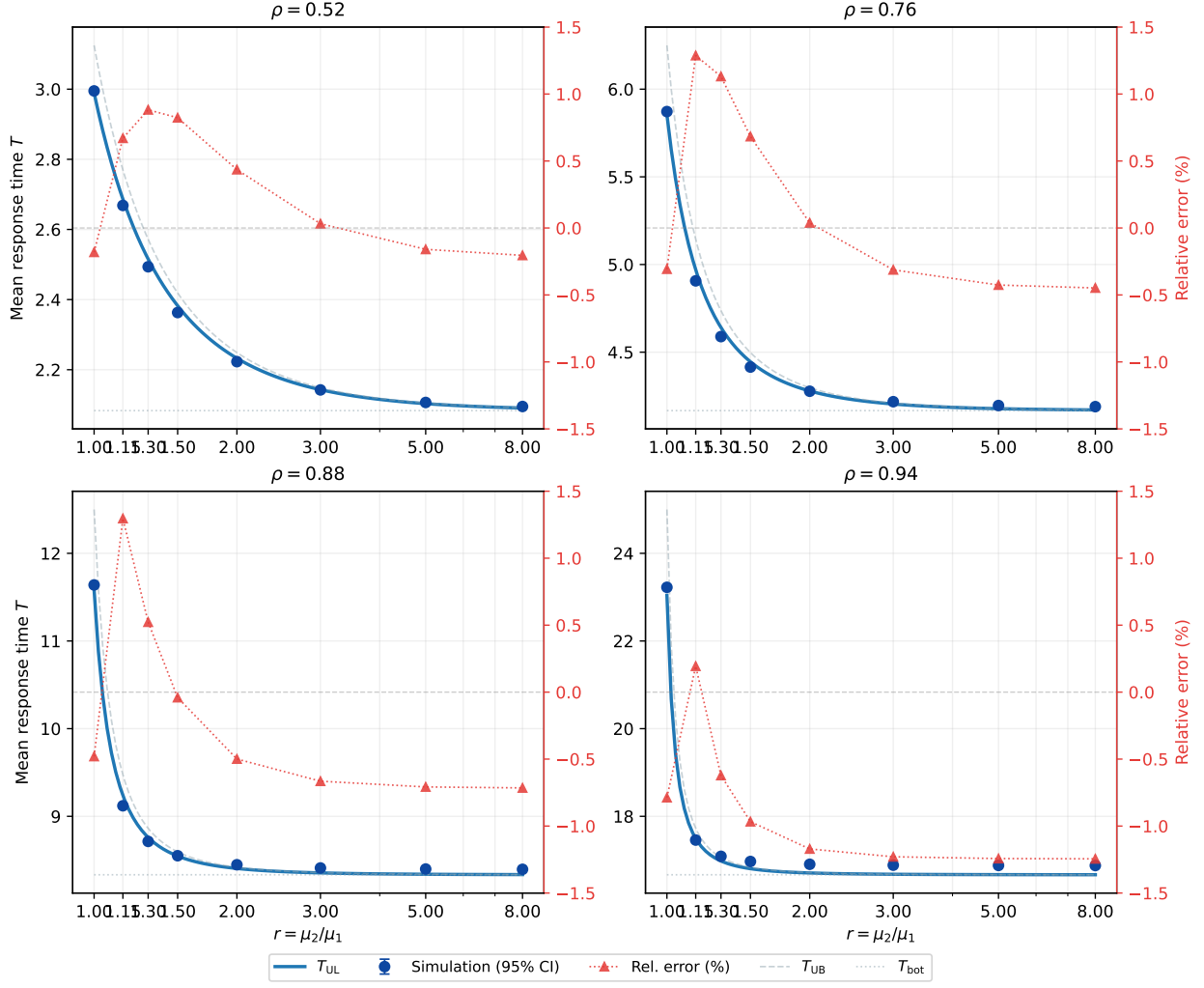


Figure 3: T_{UL} vs. heterogeneity ratio $r = \mu_{\max}/\mu_{\min}$ for four load levels $\rho \in \{0.52, 0.76, 0.88, 0.94\}$. Solid blue line: T_{UL} (analytical); filled circles: simulation with 95 % CI bars (10 M jobs, seed = 42). Dashed and dotted gray lines: upper bound T_{UB} and lower bound T_{bot} . Right axis (red triangles): relative error $(T_{UL} - T_{sim})/T_{sim} \times 100\%$.

the power-law model $h(r) = 1 + \frac{3}{8}r^{-\beta}$ is least reliable: $h(r)$ varies steeply near $r = 1$, and the resulting systematic error in the heavy-traffic anchor dominates the benefit of matching the first derivative. Both LH variants are therefore less accurate than T_{UL} (+0.73%) in this regime.

High heterogeneity ($r \geq 2$). T_{LH}^{enh} consistently outperforms T_{LH} : at $r = 2$ and $\lambda \in \{0.3, 0.6\}$ errors fall from +0.76%, +1.67% to +0.35%, +1.17%; at $r = 3$ from +1.41% to +0.54%; and at $r = 5$ from +0.71% to +0.16%. In this regime $h(r) \approx 1$ and the first light-traffic derivative $f^{(1)}(0)$ carries meaningful information about the service-time heterogeneity that improves accuracy over a wide load range.

Table 5: T_{LH} vs. $T_{\text{LH}}^{\text{enh}}$ vs. simulation for heterogeneous two-queue fork-join. T_{sim} and 95% CI as in Table 4. Errors are $(T_{\text{approx}} - T_{\text{sim}})/T_{\text{sim}} \times 100\%$.

μ_1	μ_2	λ	ρ_1	T_{sim}	$\pm 95\% \text{ CI}$	T_{LH}	Err	$T_{\text{LH}}^{\text{enh}}$	Err
1.0	1.0	0.3	0.30	2.089	0.001	2.089	+0.01%	2.089	+0.01%
1.0	1.0	0.6	0.60	3.561	0.002	3.563	+0.03%	3.563	+0.03%
1.0	1.0	0.9	0.90	13.855	0.061	13.875	+0.15%	13.875	+0.15%
1.0	1.5	0.3	0.30	1.707	0.001	1.698	−0.50%	1.710	+0.23%
1.0	1.5	0.6	0.60	2.767	0.001	2.776	+0.34%	2.801	+1.23%
1.0	1.5	0.9	0.90	10.120	0.024	10.325	+2.03%	10.362	+2.39%
1.0	2.0	0.3	0.30	1.583	0.001	1.595	+0.76%	1.589	+0.35%
1.0	2.0	0.6	0.60	2.623	0.001	2.667	+1.67%	2.654	+1.17%
1.0	2.0	0.9	0.90	10.028	0.024	10.170	+1.42%	10.150	+1.22%
1.0	3.0	0.6	0.60	2.547	0.001	2.583	+1.41%	2.561	+0.54%
1.0	5.0	0.6	0.60	2.515	0.001	2.533	+0.71%	2.519	+0.16%

6 Discussion

6.1 Relationship Between the Two Approximations

The two approximations have different theoretical foundations yet share the same two anchor points: both are exact in the homogeneous case and have the correct heavy-traffic singularity.

T_{UL} works by confining the approximation within the known bounds. Its interpolation weight $\alpha = (\rho_1 + \rho_2)/8$ has a natural interpretation: it is one-quarter of the average utilization, calibrated to reproduce the Nelson–Tantawi formula at $r = 1$. By construction, $0 \leq \alpha \leq 1/4$, so T_{UL} always lies between the bounds.

T_{LH} works from a different direction: it constructs a rational function that matches the system’s known asymptotic behavior at both traffic extremes. The heavy-traffic factor $h(r) = 1 + \frac{3}{8}r^{-\beta}$ captures the physical transition from a regime where both servers jointly saturate ($r = 1$, $h = 11/8$) to a regime where only the bottleneck server saturates ($r \rightarrow \infty$, $h \rightarrow 1$). The sharpness of this transition— $h \approx 1$ already at $r = 1.5$ —reflects the fact that even modest heterogeneity is sufficient to break the joint saturation and transfer the heavy-load correlation penalty to the bottleneck alone.

6.2 Comparison with Prior Work

Table 6 situates the two new approximations within the existing literature.

Varma and Makowski [10] applied the Reiman–Simon framework to the *symmetric* ($\mu_1 = \mu_2$) fork-join queue with $n \geq 2$ servers, showing that the light-heavy traffic interpolation recovers the exact Nelson–Tantawi formula for $n = 2$. Our T_{LH} extends this to the heterogeneous case through the $h(r)$ factor, while T_{UL} provides an alternative route via direct bound interpolation.

Table 6: Comparison of methods for fork-join mean response time ($n = 2$).

Reference	Type	Closed-form?	Exact (hom.)?	Heterogeneous?
Nelson–Tantawi [6]	Approx. (hom. only)	Yes	Yes	No
Flatto–Hahn [3, 2]	Exact (gen. func.)	No	Yes	Yes
Bacelli et al. [1]	Bounds	Partial	No	Yes
Varma–Makowski [10]	LH interp. (hom.)	Yes	Yes	No
Kemper–Mandjes [4]	Bounds + asymptotics	Partial	No	Yes
Rizk et al. [8]	Distributional bounds	No	No	Yes
Mohanty et al. [5]	Bounds ((k, k) FJ)	Partial	No	Yes
This work (T_{UL})	Bound interpolation	Yes	Yes	Yes
This work (T_{LH})	LH traffic interp.	Yes	Yes	Yes

6.3 Accuracy and Limitations

T_{UL} provides consistently small errors ($\leq 1.16\%$) and is guaranteed to lie within the bounds. Its main limitation is that the weight $\alpha = (\rho_1 + \rho_2)/8$ is derived from the homogeneous case and may not optimally represent the correlation structure at intermediate heterogeneity levels.

T_{LH} provides a tighter fit near homogeneity but has larger errors ($\leq 2.03\%$) at the transition regime ($r \approx 1.5$, heavy load), where the linear numerator has insufficient flexibility to capture the joint dependence on both r and ρ .

T_{LH}^{enh} improves on T_{LH} for $r \geq 2$ by incorporating the first light-traffic derivative, reducing errors by up to 77% in those scenarios (from $+0.71\%$ to $+0.16\%$ at $r = 5$). However, at $r \approx 1.5$ the enhancement is counterproductive: the quadratic term overshoots, yielding slightly larger errors than T_{LH} (worst case $+2.39\%$ vs. $+2.03\%$ at heavy load). The root cause is that $h(r)$ varies steeply near $r = 1$, so any error in the heavy-traffic anchor is amplified by the higher-order fit. Both LH variants are therefore less accurate than T_{UL} in the $r = 1.5$ regime. In all LH variants, the parameter β is calibrated to 10 from simulation data; an analytic determination (see Section 7) could improve performance uniformly and resolve this limitation.

Also added to Table 6 is T_{LH}^{enh} .

Table 7: Updated comparison of methods (replacing Table 6).

Reference	Type	Closed-form?	Exact (hom.)?	Heterogeneous?
Nelson–Tantawi [6]	Approx. (hom. only)	Yes	Yes	No
Flatto–Hahn [3, 2]	Exact (gen. func.)	No	Yes	Yes
Bacelli et al. [1]	Bounds	Partial	No	Yes
Varma–Makowski [10]	LH interp. (hom.)	Yes	Yes	No
Kemper–Mandjes [4]	Bounds + asymptotics	Partial	No	Yes
Rizk et al. [8]	Distributional bounds	No	No	Yes
Mohanty et al. [5]	Bounds ((k, k) FJ)	Partial	No	Yes
This work (T_{UL})	Bound interpolation	Yes	Yes	Yes
This work (T_{LH})	LH traffic interp. (0th order)	Yes	Yes	Yes
This work (T_{LH}^{enh})	LH traffic interp. (1st order)	Yes	Yes	Yes

6.4 Recommendations

For practical use:

- Use T_{UL} when a guaranteed bound is required or when high heterogeneity ($r \geq 2$) is expected. It is simpler (no free parameters) and uniformly accurate.
- Use T_{LH}^{enh} when $r \geq 2$ and the highest accuracy within the Reiman–Simon family is desired. It is strictly better than T_{LH} for $r \geq 2$ and identical to it at $r = 1$.
- Use T_{LH} (zero-order) when $r \approx 1.5$, where the first-order enhancement does not help and the simpler formula is preferable.
- Use T_{UL} and T_{LH}^{enh} together as a consistency check: agreement between them provides confidence in the estimate; significant disagreement signals a regime where further analysis (e.g., simulation or numerical inversion of the Flatto–Hahn generating function) is warranted.

7 Conclusion

We have presented three closed-form approximations for the mean response time of a heterogeneous two-queue M/M/1 fork-join system with service rates $\mu_1 \geq \mu_2$ and Poisson arrival rate λ .

The first, the *upper-lower bound interpolation*:

$$T_{UL} = \left(1 - \frac{\rho_1 + \rho_2}{8}\right) T_{UB} + \frac{\rho_1 + \rho_2}{8} T_{\text{bot}}, \quad (26)$$

is a parameter-free convex combination of the independent upper bound and the bottleneck lower bound, always within the bounds, with errors below 1.2% (validated against 20 M-job simulations with 95% confidence intervals).

The second, the *light-heavy traffic interpolation*:

$$T_{LH} = \frac{\mu_{\min} T_0 + (h(r) - \mu_{\min} T_0) \rho}{\mu_{\min} - \lambda}, \quad h(r) = 1 + \frac{3}{8} r^{-\beta}, \quad (27)$$

is grounded in the Reiman–Simon framework, matching the zeroth light-traffic derivative and a heterogeneity-adjusted heavy-traffic limit, with errors below 2.1%.

The third, the *enhanced light-heavy traffic interpolation*:

$$T_{LH}^{\text{enh}} = \frac{c_2 \rho^2 + c_1 \rho + c_0}{\mu_{\min}(1 - \rho)}, \quad (28)$$

extends T_{LH} to a quadratic numerator by additionally matching the first light-traffic derivative $f^{(1)}(0)$. It is strictly more accurate than T_{LH} for $r \geq 2$ (errors up to 77% smaller at high heterogeneity) while being identical to T_{LH} at $r = 1$. At $r \approx 1.5$, however, the enhancement is counterproductive: both LH variants are less accurate than T_{UL} in this transition regime, where the steep variation of $h(r)$ near $r = 1$ limits the Reiman–Simon approach.

All three approximations recover the Nelson–Tantawi exact formula in the homogeneous case, exhibit the correct singularity in heavy traffic, and are validated against simulation across a wide range of loads and heterogeneity levels.

Future directions include: (i) analytically determining β by deriving the slope $\partial h / \partial r|_{r=1}$ from a perturbation analysis of the Flatto–Hahn generating function, which would close the Reiman–Simon approximations without requiring simulation calibration; and (ii) extending all three methods to systems with more than two servers.

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